Type MS · Say what an algorithm (or one line) does

Answer 1 of 11 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part a.ii in this mark scheme extract

5(a)(i)	A1 A2	final score not accurate (1) because score is resetting/score keeps going back to zero (in each iteration of the loop) (1)	'Score not accurate' is not enough on its own 'Score on line 20' or 'Score output to DISPLAY' would count as final score.	**2**

Question	mp	Answer	Additional guidance	Mark
5(a)(ii)	A3 A4	Award up to **two** marks for a linked description such as: • Asking 10 addition questions (1) and calculating final score (1) • Asking randomised adding questions (1) and checking user answer against computer answer (1) • Creating adding quiz with random numbers up to 99 (1) and outputting number of correct answers (1)	Processes that can be included Concept of this being an 'Adding Quiz' required • Getting random numbers for question • Calculating sum of two random numbers for correct answers • comparing user answer to correct answer • incrementing /tracking score • asking 10 questions (loop) • outputting final score	**2**

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Say what an algorithm (or one line) does

Answer 1 of 11 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part a.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
5(b)				
	B1	Open file for writing (1)	"w" mode only Accept "w+"	
	B2	Selection used to get grade (1)	Must address all 4 grades Does not have to be accurate on boundaries e.g. >80 is ok for distinction etc.1	
	B3	Output suitable message to say file created (1)		
	B4	Evidence of 2D indexing (1)	Even if element not used in rest of program 2D indexing **or** 1D index within *for-each* structure **or** sum(1Darray) or min(1Darray) etc. At least 1 inner value accessed	
	B5	Lowest score **must** be found and omitted/subtracted from calculated scores (all arrays) (1)	Any method e.g.: • List methods • Selection statements • iterable functions must be done programmatically	
	B6	At least 1 line written to file (1)	Must contain something generated by program (total/result), does not have to be correct total or grade	
	B7	Add line feed to write line after both score and grade (1)		
	B8	Output file all correct (1)		**8**

18 of 23

Type MS · Say what an algorithm (or one line) does

Answer 1 of 11 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part a.ii in this mark scheme extract


```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

with open ("grades.txt","w") as theFile:

    # calculate each students score
    for student in Tbl_scores:
        student.sort()
        total = student[1] + student[2]

        # decide student grade
        if total >= 80:
            grade = "Distinction"
        elif total >= 65:
            grade = "Merit"
        elif total >= 50:
            grade = "Pass"
        else:
            grade = "Fail"

        # write score and grade to file
        outString = str(total) + " " + grade + "\n"
        theFile.write(outString)

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = sum(student) - min(student)

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = 0
    lowest = 51
    for score in student:
        if score < lowest:
            lowest = score
        total = total + score
    total = total - lowest

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

Type MS · Say what an algorithm (or one line) does

Answer 1 of 11 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part a.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 2 of 11 · Question 5 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

5(a)(i)	One mark for each correct row **Width** / **Length** / **Height** / **Output** 4 / 4 / 2 / Requires large container (1) 2 / 2 / 2 / Too small for cargo container (1) 3 / 8 / 5 / Too large for cargo container (1) Do not penalise spelling	3

Question Number	Answer	Additional Guidance	Mark
5(a)(ii)	Any response that indicates the concept of: Determining which size shipping container is needed for the indicated cargo.		1

Type MS · Say what an algorithm (or one line) does

Answer 2 of 11 · Question 5 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)	SEND states [numStates] TO DISPLAY (1) or SEND state TO DISPLAY (1) SET numStates TO numStates + 1 (1) or LENGTH(states) (1) Concatenation of message and variable uses " and & & numStates & TO DISPLAY (1) or & LENGTH(states) & TO DISPLAY (1) or STRING(numStates) TO DISPLAY (1) Pseudocode is correct (1)	• Accept , or + in place of & • Ignore capitalisation of commands •	4

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & numStates & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
END FOREACH
SEND "There are " & LENGTH (states) & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & STRING (numStates) TO DISPLAY
```

Type MS · Say what an algorithm (or one line) does

Answer 2 of 11 · Question 5 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Showing that index goes up to 4 and does not exceed 4 (1) Showing correspondence between values of index and loadWeight[index] (1)	• There does not have to be an explicit indication of the out-of-bounds / crashing error. • Ignore target and found columns • Allow two rows per loop with matching index and loadWeight values	2

target	found	index	loadWeight[index]
50500	FALSE	0	20000
		1	28000
		2	40000
		3	50000
		4	

target	found	index	loadWeight[index]
50500	FALSE	0	
		1	20000
		2	28000
		3	40000
		4	50000

Question Number	Answer	Additional Guidance	Mark
5(c)(ii)	1. Use of AND operator (in the test of relevant conditions) (1) 2. Use of suitable function to find the length of the array using a correct comparison to index (1) Example: WHILE ((NOT found) AND (index < LENGTH (loadWeight))) DO	• Do not penalise pseudocode syntax • Ignore changes to first condition as long as meaning is not lost	2

Type MS · Say what an algorithm (or one line) does

Answer 2 of 11 · Question 5 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 1 of 13 · find part c.vi in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 2 of 13 · find part c.vi in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 3 of 13 · find part c.vi in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 4 of 13 · find part c.vi in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 5 of 13 · find part c.vi in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 6 of 13 · find part c.vi in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 7 of 13 · find part c.vi in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 8 of 13 · find part c.vi in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 9 of 13 · find part c.vi in this mark scheme extract

Q2(c)(vi)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 10 of 13 · find part c.vi in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 11 of 13 · find part c.vi in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 12 of 13 · find part c.vi in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 3 of 11 · Question 2 part c.vi · 2 marks

2019 JUN · Paper 02 · extract 13 of 13 · find part c.vi in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 4 of 11 · Question 5 part a.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:

• Count column correct (1)
• Temp column correct (1)
• Row 2 Numbers[Count value from row above] swapped with Temp value (1)
• Row 3 Numbers[Count value from row above] swapped with Temp value (1)
• All correct (1)

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
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</table>
</td><td>Ignore any numbers that are greyed out</td><td>5</td></tr>
<tr><td>5(a)(ii)</td><td>Award one mark from:

• To reverse the contents of the array (1)
• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:

• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

Type MS · Say what an algorithm (or one line) does

Answer 4 of 11 · Question 5 part a.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Say what an algorithm (or one line) does

Answer 4 of 11 · Question 5 part a.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 5 of 11 · Question 5 part a.iii · 2 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a.iii in this mark scheme extract

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• Count column correct (1)
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Type MS · Say what an algorithm (or one line) does

Answer 5 of 11 · Question 5 part a.iii · 2 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Say what an algorithm (or one line) does

Answer 5 of 11 · Question 5 part a.iii · 2 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 1 of 13 · find part b in this mark scheme extract

4 (b)	B1 B2	**Award up to 2 marks for a linked description such as:** • To check to see if a word is the same when it is reversed / is a palindrome (1) and output an appropriate message / repaper is the same when reversed (1)		(2)

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 13 · find part b in this mark scheme extract

Coding

Question	mp	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 3 of 13 · find part b in this mark scheme extract

4 (a)	**Award one mark for each of:**			
		Validation		
	A1	Suitable input prompt and storing it in binaryPattern (1)		
	A2	Length of binary pattern validated as 8 (1)		
	A3	Loop used to check the length of binary pattern (1)		
		Binary to denary		
	A4	Loop through each digit in the binary pattern (1)		
	A5	Correct conversion of at least one binary digit to relevant placeholder (1)		
	A6	Attempt at running total using denaryNumber (1)		
	A7	Correct running total using denaryNumber (1)		
	A8	Print statement includes the binary pattern and the denary number (1)		
	A9	Print statement outside of conversion loop (1)		
	A10	Comment explains how conversion works (1)		
	A11	Program executes correctly for any binary pattern that is exactly 8 characters long only (1)		(11)
Code examples				

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 4 of 13 · find part b in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 5 of 13 · find part b in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 6 of 13 · find part b in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

Question	mp	Answer	Additional Guidance	Mark

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 7 of 13 · find part b in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 8 of 13 · find part b in this mark scheme extract



```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 9 of 13 · find part b in this mark scheme extract

Java

```java
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 10 of 13 · find part b in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 11 of 13 · find part b in this mark scheme extract

Python

```python
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

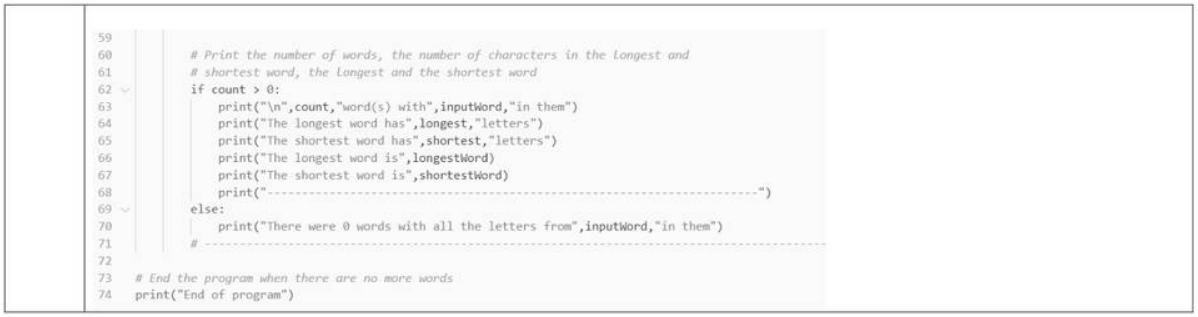
                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 12 of 13 · find part b in this mark scheme extract



```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73      # End the program when there are no more words
74      print("End of program")
```

Type MS · Say what an algorithm (or one line) does

Answer 6 of 11 · Question 4 part b · 2 marks

2021 NOV · Paper 02 · extract 13 of 13 · find part b in this mark scheme extract

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Type MS · Say what an algorithm (or one line) does

Answer 7 of 11 · Question 6 part b · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

6(a)	Award **one** mark for each of: Method • Flowchart (1) Justification • It is a visual representation / does not use many words / does not rely on use of English language (1) • It does not rely on understanding specific syntax (1) • Overview without unnecessary detail (1) OR Method • Pseudocode/structured English/written description (1) Justification • It does not rely on understanding specific syntax (1) • It uses English language words (1)	Valid method must be specified first before mark for justification can be awarded.	2

Type MS · Say what an algorithm (or one line) does

Answer 7 of 11 · Question 6 part b · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for any of the following: • The test on line 3 has already limited the values to 1 and 2 (1) • The only value left for inNum to be (on line 7) is 2 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award up to **two** marks for a linked description, such as: • The oldIndex value will go from highest to lowest (1) the step value is negative/decremented by 1 (each iteration) (1)	Allow 4 to 0 instead of highest and lowest	2

Question Number	Answer	Additional Guidance	Mark
6(c)(ii)	Award **one** mark for any of the following: • Reverse the (array of) scores (1) • Copy the oldScores into newScores, in reverse order (1)		1

Type MS · Say what an algorithm (or one line) does

Answer 7 of 11 · Question 6 part b · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Question Number	Indicative Content	Additional Guidance	Mark
6(d)	**Growth** • Higher bandwidth available both on wired and wireless networks • Reduced costs of large capacity storage devices • Improved levels of encryption / authentication • Increased trust in service providers • Growth of service provider options • Increased business models that aid scalability regarding storage capacity and costs **Benefits** • Sharing / collaboration on documents and setting of access permissions • Access from any location connected to a WAN • Reduced cost of local computing devices with less storage required • Built in backups of data on server-based systems **Drawbacks** • Increased security risks incurred from outsourcing a service • Relies on high-speed network connections • Reliance on service providers continuing to provide a high level of service • Increased chances of external data breaches • Identifying responsibilities for data breaches / legal jurisdiction • Increased power consumption from server systems and associated cooling systems • Increased use of networks and network traffic which will have an impact on other network users **Conclusion** • A conclusion does not have to be stated explicitly, but can be inferred from the discussion		6

Type MS · Say what an algorithm (or one line) does

Answer 7 of 11 · Question 6 part b · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Say what an algorithm (or one line) does

Answer 7 of 11 · Question 6 part b · 1 mark

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Type MS · Say what an algorithm (or one line) does

Answer 8 of 11 · Question 6 part c.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

6(a)	Award **one** mark for each of: Method • Flowchart (1) Justification • It is a visual representation / does not use many words / does not rely on use of English language (1) • It does not rely on understanding specific syntax (1) • Overview without unnecessary detail (1) OR Method • Pseudocode/structured English/written description (1) Justification • It does not rely on understanding specific syntax (1) • It uses English language words (1)	Valid method must be specified first before mark for justification can be awarded.	2

Type MS · Say what an algorithm (or one line) does

Answer 8 of 11 · Question 6 part c.i · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for any of the following: • The test on line 3 has already limited the values to 1 and 2 (1) • The only value left for inNum to be (on line 7) is 2 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award up to **two** marks for a linked description, such as: • The oldIndex value will go from highest to lowest (1) the step value is negative/decremented by 1 (each iteration) (1)	Allow 4 to 0 instead of highest and lowest	2

Question Number	Answer	Additional Guidance	Mark
6(c)(ii)	Award **one** mark for any of the following: • Reverse the (array of) scores (1) • Copy the oldScores into newScores, in reverse order (1)		1

Type MS · Say what an algorithm (or one line) does

Answer 8 of 11 · Question 6 part c.i · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

Question Number	Indicative Content	Additional Guidance	Mark
6(d)	**Growth** • Higher bandwidth available both on wired and wireless networks • Reduced costs of large capacity storage devices • Improved levels of encryption / authentication • Increased trust in service providers • Growth of service provider options • Increased business models that aid scalability regarding storage capacity and costs **Benefits** • Sharing / collaboration on documents and setting of access permissions • Access from any location connected to a WAN • Reduced cost of local computing devices with less storage required • Built in backups of data on server-based systems **Drawbacks** • Increased security risks incurred from outsourcing a service • Relies on high-speed network connections • Reliance on service providers continuing to provide a high level of service • Increased chances of external data breaches • Identifying responsibilities for data breaches / legal jurisdiction • Increased power consumption from server systems and associated cooling systems • Increased use of networks and network traffic which will have an impact on other network users **Conclusion** • A conclusion does not have to be stated explicitly, but can be inferred from the discussion		6

Type MS · Say what an algorithm (or one line) does

Answer 8 of 11 · Question 6 part c.i · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
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Type MS · Say what an algorithm (or one line) does

Answer 8 of 11 · Question 6 part c.i · 2 marks

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Type MS · Say what an algorithm (or one line) does

Answer 9 of 11 · Question 6 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

6(a)	Award **one** mark for each of: Method • Flowchart (1) Justification • It is a visual representation / does not use many words / does not rely on use of English language (1) • It does not rely on understanding specific syntax (1) • Overview without unnecessary detail (1) OR Method • Pseudocode/structured English/written description (1) Justification • It does not rely on understanding specific syntax (1) • It uses English language words (1)	Valid method must be specified first before mark for justification can be awarded.	2

Type MS · Say what an algorithm (or one line) does

Answer 9 of 11 · Question 6 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for any of the following: • The test on line 3 has already limited the values to 1 and 2 (1) • The only value left for inNum to be (on line 7) is 2 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award up to **two** marks for a linked description, such as: • The oldIndex value will go from highest to lowest (1) the step value is negative/decremented by 1 (each iteration) (1)	Allow 4 to 0 instead of highest and lowest	2

Question Number	Answer	Additional Guidance	Mark
6(c)(ii)	Award **one** mark for any of the following: • Reverse the (array of) scores (1) • Copy the oldScores into newScores, in reverse order (1)		1

Type MS · Say what an algorithm (or one line) does

Answer 9 of 11 · Question 6 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

Question Number	Indicative Content	Additional Guidance	Mark
6(d)	**Growth** • Higher bandwidth available both on wired and wireless networks • Reduced costs of large capacity storage devices • Improved levels of encryption / authentication • Increased trust in service providers • Growth of service provider options • Increased business models that aid scalability regarding storage capacity and costs **Benefits** • Sharing / collaboration on documents and setting of access permissions • Access from any location connected to a WAN • Reduced cost of local computing devices with less storage required • Built in backups of data on server-based systems **Drawbacks** • Increased security risks incurred from outsourcing a service • Relies on high-speed network connections • Reliance on service providers continuing to provide a high level of service • Increased chances of external data breaches • Identifying responsibilities for data breaches / legal jurisdiction • Increased power consumption from server systems and associated cooling systems • Increased use of networks and network traffic which will have an impact on other network users **Conclusion** • A conclusion does not have to be stated explicitly, but can be inferred from the discussion		6

Type MS · Say what an algorithm (or one line) does

Answer 9 of 11 · Question 6 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Say what an algorithm (or one line) does

Answer 9 of 11 · Question 6 part c.ii · 1 mark

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Type MS · Say what an algorithm (or one line) does

Answer 10 of 11 · Question 6 part c · 1 mark

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6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Say what an algorithm (or one line) does

Answer 10 of 11 · Question 6 part c · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

6(b)(iv)	Award **one** mark for:
	
	• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)
	`LENGTH(arrival) - (late + early)`
		Allow number of trains to be hard coded as 7 in place of
		`LENGTH(arrival)`
		
		Allow equivalent expressions:
		`LENGTH(arrival) - late - early`
			1

6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

Type MS · Say what an algorithm (or one line) does

Answer 10 of 11 · Question 6 part c · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

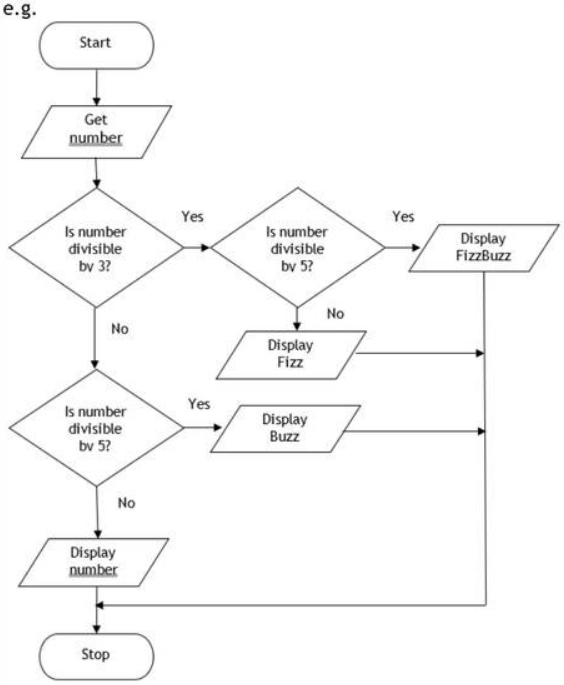
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Say what an algorithm (or one line) does

Answer 10 of 11 · Question 6 part c · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

Type MS · Say what an algorithm (or one line) does

Answer 10 of 11 · Question 6 part c · 1 mark

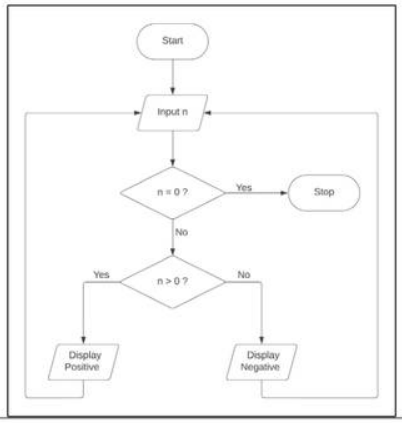
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Type MS · Say what an algorithm (or one line) does

Answer 11 of 11 · Question 6 part b.iii · 1 mark

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Question	Answer	Additional Guidance	mark
6(a)	Award **one** mark for each of: • Start and Input n identified (accept number or num) (1) • Decision box test for n = 0/n<>0 with yes & no/true & false labels that correctly lead to End/Stop (1) • Decision box test for n > 0 or n<0 with yes & no/true & false labels (1) • Correct messages output including leads from correct yes & no/true & false (1) • Flowchart is fully connected and functionally correct (1) • Correct flowchart symbols used throughout for terminators, input/output and decision boxes (1) Example: 	Accept == for MP2 For MP2 and MP3 decision boxes must have yes/no or true/false labels. For MP5 there must be arrows on connections	6

Type MS · Say what an algorithm (or one line) does

Answer 11 of 11 · Question 6 part b.iii · 1 mark

2025 JUN · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(i)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>7</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td><td></td></tr><tr><td></td><td></td><td>4</td><td></td></tr><tr><td></td><td></td><td>5</td><td></td></tr><tr><td></td><td></td><td>6</td><td></td></tr><tr><td></td><td>TRUE</td><td>7</td><td></td></tr></table></td><td>Accept for first pair
<table><tr><th>number</th><th>flag</th></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>TRUE</td></tr></table>Do not award either mark point if first row is 7 FALSE 2</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(ii)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>9</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td>TRUE</td><td>3</td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table></td><td>Accept for first pair
<table><tr><th>number</th><th>flag</th></tr><tr><td>9</td><td>FALSE</td></tr><tr><td>9</td><td>FALSE</td></tr><tr><td>9</td><td>TRUE</td></tr></table>Do not award either mark point if first row is 9 FALSE 2</td><td>2</td></tr>
</table>

Type MS · Say what an algorithm (or one line) does

Answer 11 of 11 · Question 6 part b.iii · 1 mark

2025 JUN · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
6(b)(iii)	Award **one** mark for: • To determine if a number is a prime number/none of the test numbers divide exactly into it (1)	Ignore reference to display the number	1
Question	Answer	Additional Guidance	Mark
6(b)(iv)	Award up to **two** marks for a linked explanation such as: • It makes the algorithm more efficient (1) because the loop will terminate as soon as a test number divides exactly into the number entered (1) • A count-controlled loop could make more comparisons/be less efficient (1) because it would not terminate as soon as the test number divides exactly into the number entered (1) • A count-controlled loop executes a fixed number of times/even after condition is met (1) post-condition loop stops when condition is met (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 6	**13**

Type MS · Say what an algorithm (or one line) does

Answer 11 of 11 · Question 6 part b.iii · 1 mark

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